

Hospital Management & Appointment System

MediDesk — Final Project Review

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Problem & Objectives

The problem

Manual paperwork and long waiting times

Inconsistent appointment booking and tracking

Data-entry errors, hard-to-reach patient records

Billing, pharmacy and lab run in silos

Our objectives

Centralise operations on one web platform

Simplify registration and appointment booking

Fast EMR access for doctors

Cut waiting time with token & queue management

Link billing, pharmacy and lab to patient records

Modules & Functionalities

Patient Registration

Profiles & history

Appointment Booking

Based on doctor availability

Doctor Schedules

Slots & availability

Electronic Records

Diagnosis, prescriptions

Token & Queue

Live queue status

Billing & Insurance

Bills, payments, coverage

Pharmacy & Inventory

Stock, expiry, suppliers

Laboratory Reports

Tests & report files

Admin Dashboard

Hospital-wide view

Role-based Access

Patient · Doctor · Admin

Scope: a single hospital / clinic with patient, doctor and admin roles.

System Architecture & Technology

USER LAYER Patient · Doctor · Admin

FRONTEND React.js (Vite) — three role-based portals

BACKEND Node.js + Express.js — REST APIs, JWT auth, routes · controllers · models

DATABASE MySQL + Sequelize — hospital_management schema

EXTERNAL Twilio SMS notifications

Toolchain

React.js

Node.js &
Express.js

MySQL &
Sequelize

JWT

Postman

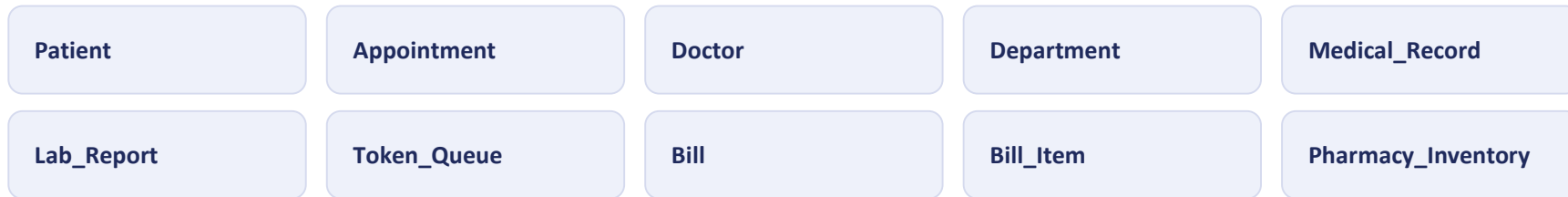
Git & GitHub

VS Code · npm

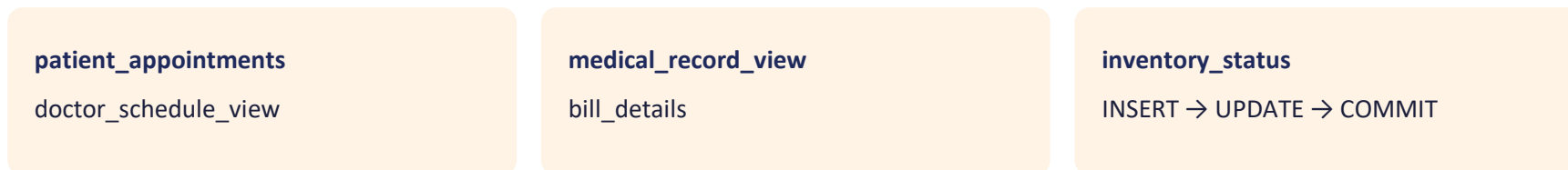
Database Design

Normalised ER model with primary/foreign key constraints linking every entity.

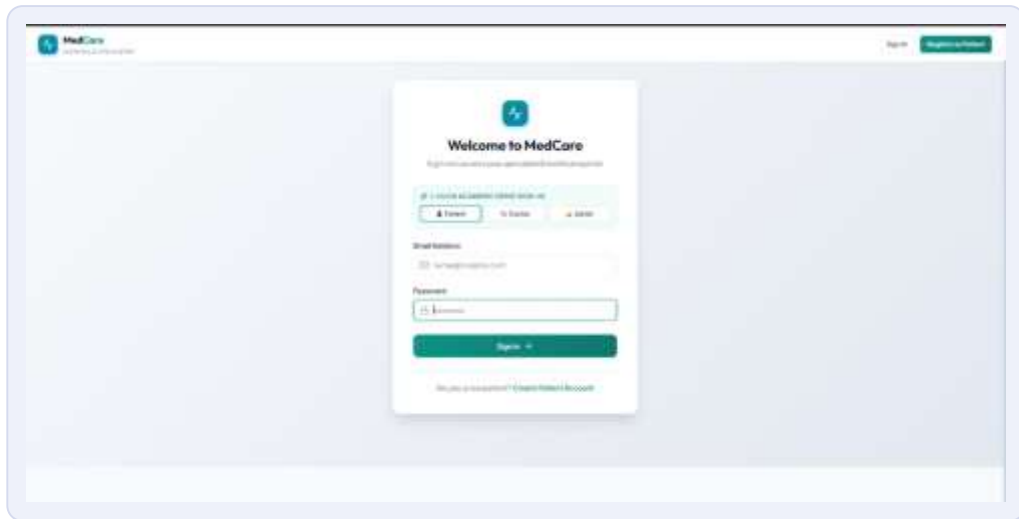
Core entities



Views & transactions



Implementation — Secure Access



Login page with one-click demo sign-in for each role

JWT authentication

Every request is verified; portals open by role

Role-based access

Patient, Doctor and Admin see only their own modules

Modular backend

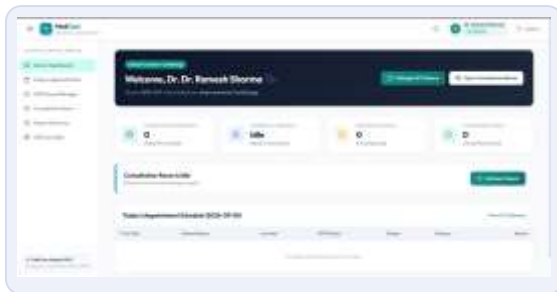
Express routes, controllers and models over MySQL with input validation

Implementation — Role-based Portals



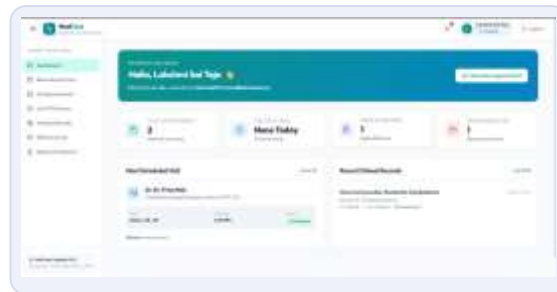
Admin portal

Command center with patient, doctor, revenue and OPD queue KPIs, inventory alerts and appointment monitoring.



Doctor portal

Clinical console with today's appointments, OPD queue manager, consultation room and patient directory.



Patient portal

Book appointments, live OPD token, medical records, bills and diagnostic reports in one place.

Conclusion & Future Work

Achieved

One connected platform for patients, doctors and admins

Role-based portals with dashboards

Normalised schema with views and verified transactions

Secure, modular REST backend

Future work

SMS notifications via Twilio

Analytics dashboard

Telemedicine / video consultation

Docker deployment

Thank you · P. Rahul (2520090016) · A. Lakshmi Sai Teja (2520080005)